

ECON 1550: International Finance

Money, Interest Rates, and Exchange Rates

A model of the money market

Exogenous variables		Endogenous variables			
Variable	Description	Variable	Description	Equation	Type of equation
Y	Real income	R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^s	Money supply	M^d	Money demand	$M^d = M^s$	Equilibrium condition
P	Price level				

Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
 - Can be used for transactions
 - Pays no interest
- Bonds
 - Cannot be used for transactions
 - Pay interest $R \geq 0$

Money Demand

- Money demand is higher when:
 - Higher price level $P \rightarrow$ need more money to buy goods
 - Lower nominal interest rate $R \rightarrow$ bonds less attractive
 - Higher income $Y \rightarrow$ want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L(\underset{(-)}{R}, \underset{(+)}{Y})$$

Real money demand

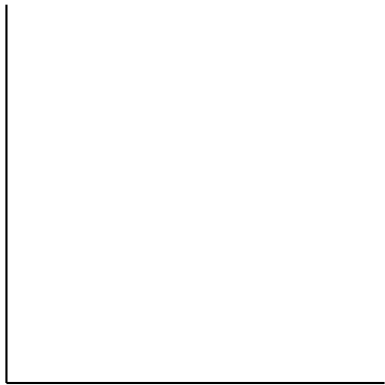
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

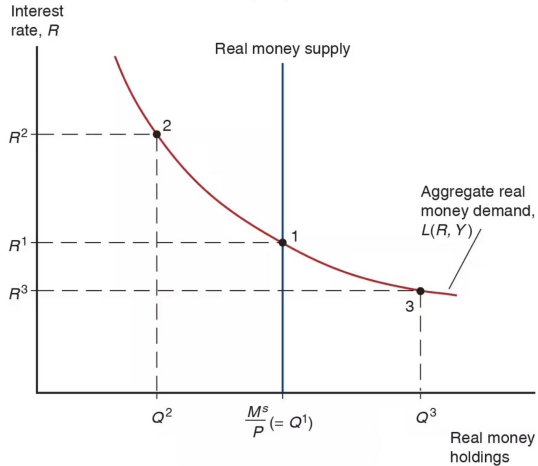
and real money supply

$$\frac{M^s}{P}$$

Equilibrium in money market



Equilibrium in money market



Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

Many types of money (1/2)

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged for a commodity

Many types of money (2/2)

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

Assets	Liabilities	Assets	Liabilities	Assets	Liabilities
\$0	\$0	\$1 currency	\$1 money	\$1 bonds	\$1 money
$t = 0$		$t = 1$		$t = 2$	

Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

Short-run FX and money market model

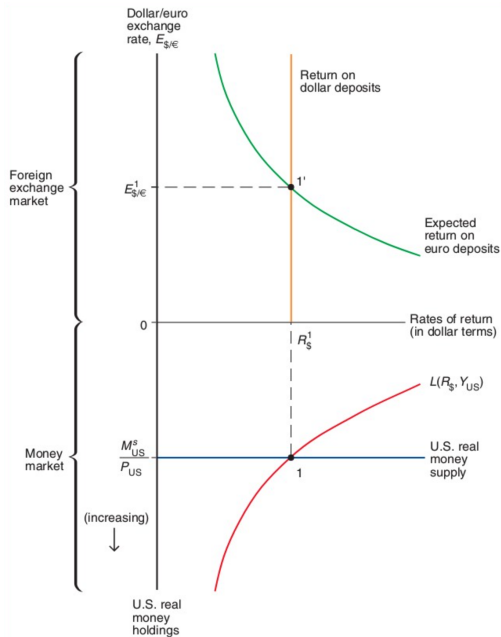
Exogenous variables

Variable	Description
R^*	Foreign interest rate
E^e	Expected exchange rate
Y	Real income
M^s	Money supply
P	Price level

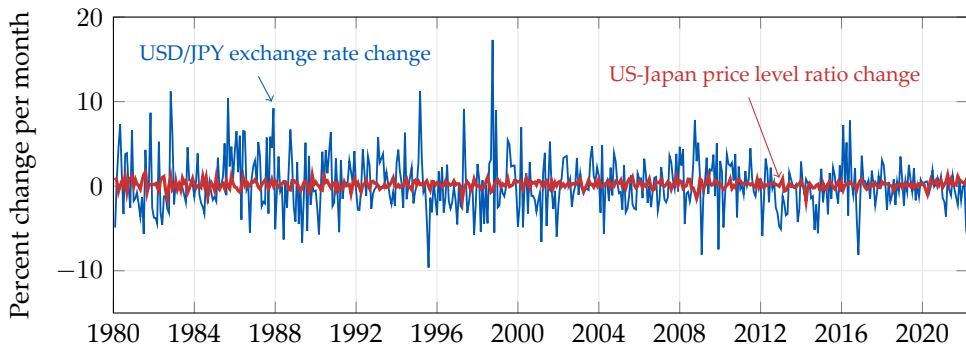
Endogenous variables

Variable	Description	Equation	Type of equation
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Equilibrium condition
R	Domestic interest rate	$M^d/P = L(R, Y)$	Behavioral equation
M^d	Money demand	$M^d = M^s$	Equilibrium condition

Short-run equilibrium in FX and money markets



Exchange rates are very volatile



Source: [FRED/CPIAUCSL](#); [FRED/CPALCY01JPM661N](#); [FRED/DEXJPUS](#) (end-of-month, inverted to USD/JPY).

Conceptual definition of long-run equilibrium

- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

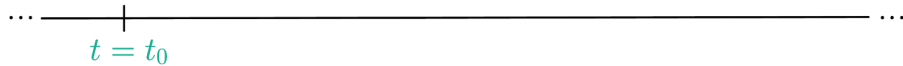
Definition of the long run in the model

1. In the long run, $E = E^e$
 - By definition of the long run, all variables constant
 - By assumption of rational expectations, expectations must be correct in the long run
 - In the model, it means $E_{LR} = E_{LR}^e$

Assumptions about the short run

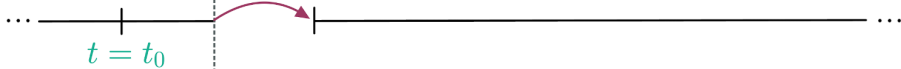
1. In the short run, P is fixed
 - Because prices are “sticky”
 - In the model, it means $P_0 = P_{SR}$
2. In the short run, E^e equals long-run E
 - By assumption of rational expectations
 - In the model, it means $E_{SR}^e = E_{LR}$

Initial long run
equilibrium

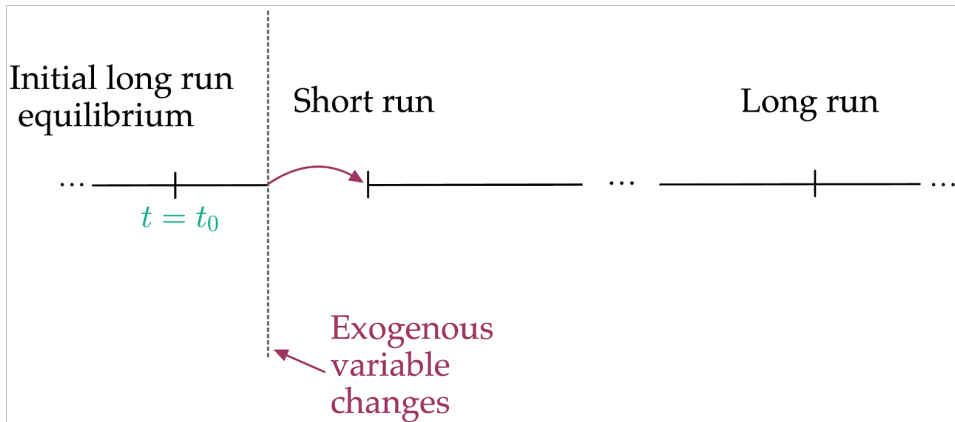


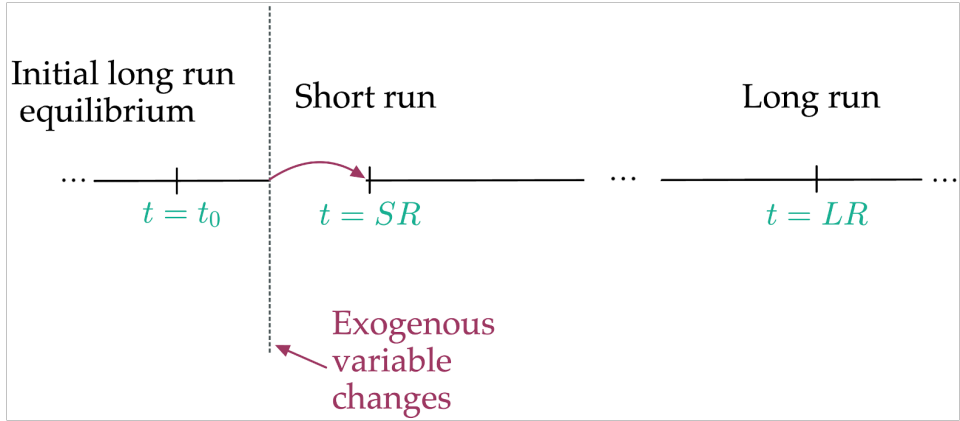
Initial long run
equilibrium

Short run



Exogenous
variable
changes





Results in the long run (1/3)

1. Long-run interest rate R_{LR} from FX market
 - Using $E_{LR} = E_{LR}^e$ and UIP:

$$R_{LR} = R^* + \frac{E_{LR}^e}{E_{LR}} - 1$$

$$\Rightarrow R_{LR} = R^*$$

Results in the long run (2/3)

2. Long-run price level P_{LR} from the money market
- Using $R_{LR} = R^*$ and money market equilibrium condition:

$$\frac{M^s}{P_{LR}} = L(R_{LR}, Y) = L(R^*, Y)$$
$$\Rightarrow P_{LR} = \frac{M^s}{L(R^*, Y)}$$

Results in the long run (3/3)

3. Long-run exchange rate E_{LR} from PPP (purchasing power parity)
 - Because E_{LR} is a nominal price, in the long run it moves proportionally to the price level:

$$E_{LR} \propto P_{LR}$$

Results in the short run (1/2)

1. Short-run exchange rate E_{SR} from UIP with $E^e = E_{LR}$
 - Using UIP and the short-run assumptions:

$$R = R^* + \frac{E_{LR} - E_{SR}}{E_{SR}} \quad \Rightarrow \quad E_{SR} = \frac{E_{LR}}{1 + R - R^*}$$

Results in the short run (2/2)

2. Short-run domestic interest rate R_{SR} from the money market

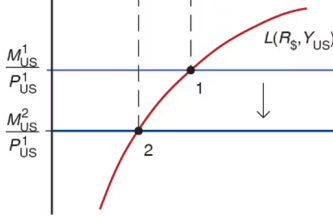
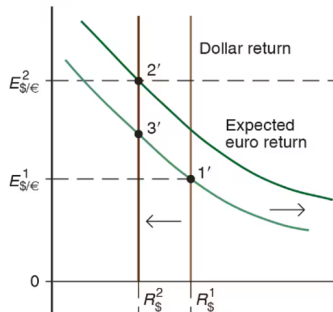
- Using money market equilibrium condition and P fixed at P_0 in the short run:

$$\frac{M^s}{P_{SR}} = L(R_{SR}, Y) \quad \Rightarrow \quad \frac{M^s}{P_0} = L(R_{SR}, Y),$$

which can be solved for R_{SR}

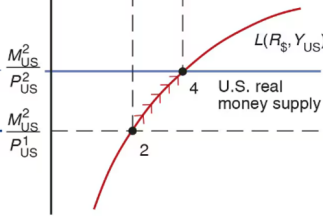
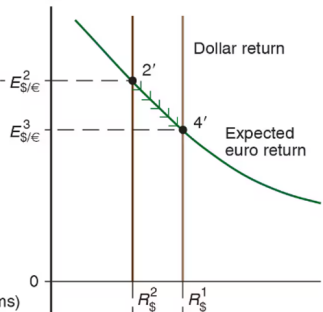
Permanent Money Supply Changes

Dollar/euro exchange
rate, $E_{\$/\epsilon}$



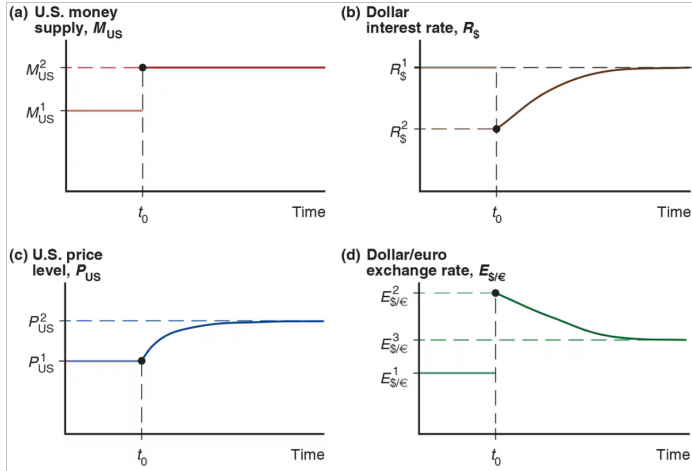
U.S. real
money holdings

$E_{\$/\epsilon}$



U.S. real
money holdings

Exchange Rate Overshooting



Long-run FX and money market model

Exogenous variables

Variable	Description
Y^n	Potential output
M^s	Money supply
R^*	Foreign interest rate

Endogenous variables

Variable	Description	Equation	Type of eq
E	Exchange rate	$R = R^* + \frac{E^e - E}{E}$	Eq. condition
M^d	Money demand	$M^d/P = M^s/P$	Eq. condition
E^e	Expected exchange rate	$E^e = E$	Behavioral
R	Domestic interest rate	$P = \frac{M^d}{L(R, Y^n)}$	Eq. condition
P	Price level	P fixed in the short run	Behavioral

Overshooting

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1. Properties of initial, SR, LR equilibria

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2. Dynamic FX and money market model

Overshooting

1. Properties of initial, SR, LR equilibria
2. Dynamic FX and money market model
3. Example
 - Solve with equations
 - Solve graphically

Properties of initial, SR, and LR equilibria

Properties of initial, SR, and LR equilibria

Long run:

Properties of initial, SR, and LR equilibria

Long run:

- $E^e = E$

Properties of initial, SR, and LR equilibria

Long run:

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$$\Rightarrow E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR}$$

Properties of initial, SR, and LR equilibria

Long run:

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- E proportional to P and M^s

Properties of initial, SR, and LR equilibria

Long run:

- $E^e = E$

$$\Rightarrow E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR}$$

- E proportional to P and M^s

$$\Rightarrow E_0 = kM_0^s \text{ and } E_{LR} = kM_{LR}^s \quad (k \text{ is a parameter})$$

Properties of initial, SR, and LR equilibria

Short run:

Properties of initial, SR, and LR equilibria

Short run:

- Prices are fixed

Properties of initial, SR, and LR equilibria

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Properties of initial, SR, and LR equilibria

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$$\Rightarrow P_{SR} = P_0$$

- Rational expectations

$$\Rightarrow E_{SR}^e = E_{LR}$$

Transition between SR and LR

Transition between SR and LR

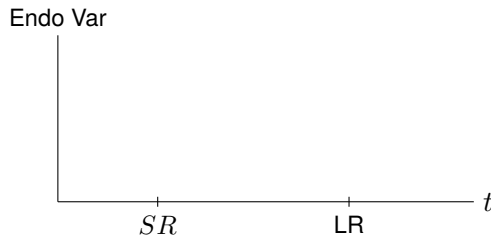
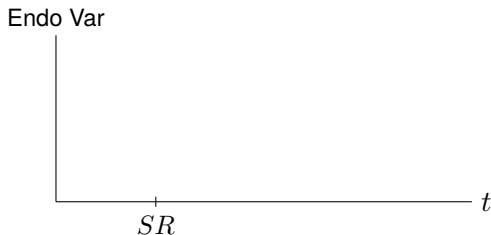
- Slow monotonic transition between SR value and LR value

Transition between SR and LR

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- No indication of shape

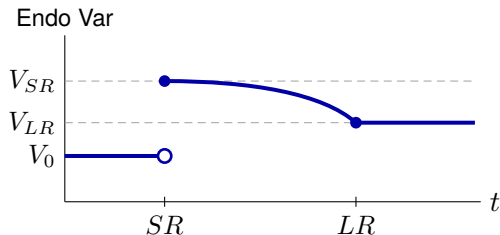
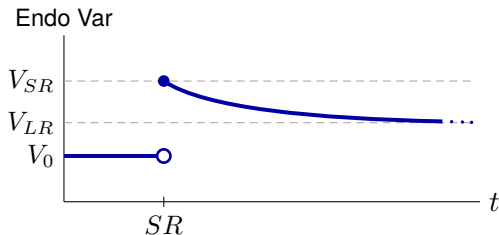
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Dynamic FX and money market model

Dynamic FX and money market model

Exogenous variables

Variable	Description
M^s	Path of money supply
R^*	Path of foreign interest rate
Y	Path of income (GNP)
k	Long-run proportionality constant for E and M^s

Dynamic FX and money market model

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P	Price level
R	Domestic interest rate
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Equations that apply at all times

$$\text{(UIP)} : R = R^* + \frac{E^e}{E} - 1 \quad \text{(equilibrium condition)}$$

$$\text{(MD)} : \frac{M^d}{P} = L \left(\underset{(-)}{R}, \underset{(+)}{Y} \right) \quad \text{(behavioral)}$$

$$\text{(MS = MD)} : \frac{M^s}{P} = \frac{M^d}{P} \quad \text{(equilibrium condition)}$$

Equations that apply in SR equilibria only

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(Sticky P) : $P_{SR} = P_0$

(behavioral)

Equations that apply in SR equilibria only

(Sticky P) : $P_{SR} = P_0$ (behavioral)

(RE) : $E_{SR}^e = E_{LR}$ (behavioral)

Equations that apply in LR equilibria only

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$$\text{(PPP) or (Flex P)} : E_0 = kM_0^s, \quad E_{LR} = kM_{LR}^s \quad \text{(behavioral)}$$

Equations that apply in LR equilibria only

(PPP) or (Flex P) : $E_0 = kM_0^s, \quad E_{LR} = kM_{LR}^s$ (behavioral)

(Defn LR) or (RE) : $E_0^e = E_0, \quad E_{LR}^e = E_{LR}$ (behavioral)

Example

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Need to specify:

1. Function $L(R, Y)$

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1. $L(R, Y) = \frac{Y}{1 + R}$
2. $R^* = 0.08, Y = 1, k = 1$

Example

Need to specify:

1. Function $L(R, Y)$
2. Path of exogenous variables

Assume:

1. $L(R, Y) = \frac{Y}{1 + R}$

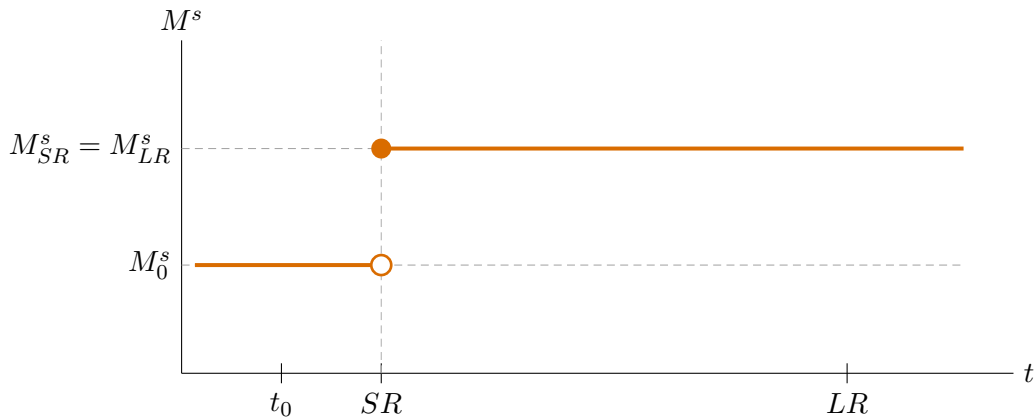
2. $R^* = 0.08, Y = 1, k = 1$

$$\{M_0^s, M_{SR}^s, M_{LR}^s\} = \{1, 1.05, 1.05\}$$

Example: Permanent increase in M^s



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Solution steps

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2. Solve LR equilibrium ($t = LR$)
3. Solve SR equilibrium ($t = SR$)

Solution sub-steps

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LR (steps 1 and 2)

(a) Flex P / PPP $\longrightarrow E$

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(c) UIP $\longrightarrow R$

Solution sub-steps

LR (steps 1 and 2)

(a) Flex P / PPP $\longrightarrow E$

(b) Defn LR / RE $\longrightarrow E^e$

(c) UIP $\longrightarrow R$

(d) MS = MD $\longrightarrow P$

Solution sub-steps

LR (steps 1 and 2)

(a) Flex P / PPP $\longrightarrow E$

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SR (step 3)

(a) P fixed $\longrightarrow P$

Solution sub-steps

LR (steps 1 and 2)

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Solution sub-steps

LR (steps 1 and 2)

- (a) Flex P / PPP $\longrightarrow E$
- (b) Defn LR / RE $\longrightarrow E^e$
- (c) UIP $\longrightarrow R$
- (d) MS = MD $\longrightarrow P$

SR (step 3)

- (a) P fixed $\longrightarrow P$
- (b) MS = MD $\longrightarrow R$
- (c) RE $\longrightarrow E^e$

Solution sub-steps

LR (steps 1 and 2)

- (a) Flex P / PPP $\longrightarrow E$
- (b) Defn LR / RE $\longrightarrow E^e$
- (c) UIP $\longrightarrow R$
- (d) MS = MD $\longrightarrow P$

SR (step 3)

- (a) P fixed $\longrightarrow P$
- (b) MS = MD $\longrightarrow R$
- (c) RE $\longrightarrow E^e$
- (d) UIP $\longrightarrow E$

Step 1: Solve initial LR equilibrium ($t = t_0$)

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Step 1: Solve initial LR equilibrium ($t = t_0$)

$$(a) \quad \text{Flex P / PPP} \rightarrow E \quad : \quad E_0 = kM_0^s = 1 \cdot 1 = 1$$

$$(b) \quad \text{Defn LR / RE} \rightarrow E^e \quad : \quad E_0^e = E_0 = 1$$

$$(c) \quad \text{UIP} \rightarrow R \quad : \quad R_0 = R^* + \frac{E_0^e}{E_0} - 1 \\ = 0.08 + \frac{1}{1} - 1 = 0.08$$

Step 1: Solve initial LR equilibrium ($t = t_0$)

$$\begin{aligned} \text{(d) } MS = MD \rightarrow P & \quad : \quad \frac{M_0^s}{P_0} = \frac{Y_0}{1 + R_0} \\ & \Rightarrow P_0 = M_0^s \cdot \frac{1 + R_0}{Y_0} \\ & \quad \quad \quad = 1 \times 1.08 = 1.08 \end{aligned}$$

Step 1: Solve initial LR equilibrium ($t = t_0$)

$$\begin{aligned} \text{(d) } MS=MD \rightarrow P & \quad : \quad \frac{M_0^s}{P_0} = \frac{Y_0}{1 + R_0} \\ & \Rightarrow P_0 = M_0^s \cdot \frac{1 + R_0}{Y_0} \\ & \quad = 1 \times 1.08 = 1.08 \end{aligned}$$

Solution for t_0 : $P_0 = 1.08$, $R_0 = 0.08$, $E_0^e = 1$, $E_0 = 1$

Step 2: Solve LR equilibrium ($t = LR$)

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(a) Flex P / PPP $\rightarrow E$: $E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$

Step 2: Solve LR equilibrium ($t = LR$)

(a) Flex P / PPP $\rightarrow E$: $E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$

(b) Defn LR / RE $\rightarrow E^e$: $E_{LR}^e = E_{LR} = 1.05$

Step 2: Solve LR equilibrium ($t = LR$)

(a) Flex P / PPP $\rightarrow E$: $E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$

(b) Defn LR / RE $\rightarrow E^e$: $E_{LR}^e = E_{LR} = 1.05$

(c) UIP $\rightarrow R$: $R_{LR} = R_{LR}^* + \frac{E_{LR}^e}{E_{LR}} - 1$
 $= 0.08 + \frac{1.05}{1.05} - 1 = 0.08$

Step 2: Solve LR equilibrium ($t = LR$)

$$\begin{aligned} \text{(d) } MS=MD \rightarrow P & \quad : \quad \frac{M_{LR}^s}{P_{LR}} = \frac{Y_{LR}}{1 + R_{LR}} \\ & \Rightarrow P_{LR} = M_{LR}^s \cdot \frac{1 + R_{LR}}{Y_{LR}} \\ & \quad = 1.05 \times 1.08 = 1.134 \end{aligned}$$

Step 2: Solve LR equilibrium ($t = LR$)

$$\begin{aligned} \text{(d) MS=MD} \rightarrow P & : \frac{M_{LR}^s}{P_{LR}} = \frac{Y_{LR}}{1 + R_{LR}} \\ & \Rightarrow P_{LR} = M_{LR}^s \cdot \frac{1 + R_{LR}}{Y_{LR}} \\ & = 1.05 \times 1.08 = 1.134 \end{aligned}$$

Solution for LR: $P_{LR} = 1.134$, $R_{LR} = 0.08$, $E_{LR}^e = 1.05$, $E_{LR} = 1.05$

Step 3: Solve SR equilibrium ($t = SR$)

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(a) P fixed $\rightarrow P$: $P_{SR} = P_0 = 1.08$

Step 3: Solve SR equilibrium ($t = SR$)

$$(a) \quad P \text{ fixed} \rightarrow P \quad : \quad P_{SR} = P_0 = 1.08$$

$$(b) \quad MS=MD \rightarrow R \quad : \quad \frac{M_{SR}^s}{P_{SR}} = \frac{Y_{SR}}{1 + R_{SR}}$$

$$\Rightarrow \frac{1.05}{1.08} = \frac{1}{1 + R_{SR}}$$

$$\Rightarrow 1 + R_{SR} = \frac{1.08}{1.05}$$

$$\Rightarrow R_{SR} = \frac{1.08}{1.05} - 1 \approx 0.02857$$

Step 3: Solve SR equilibrium ($t = SR$)

(c) $\text{RE} \rightarrow E^e$: $E_{SR}^e = E_{LR} = 1.05$

Step 3: Solve SR equilibrium ($t = SR$)

$$(c) \quad RE \rightarrow E^e : \quad E_{SR}^e = E_{LR} = 1.05$$

$$(d) \quad UIP \rightarrow E : \quad R_{SR} = R_{SR}^* + \frac{E_{SR}^e}{E_{SR}} - 1$$

$$\Rightarrow 0.02857 = 0.08 + \frac{1.05}{E_{SR}} - 1$$

$$\Rightarrow E_{SR} = \frac{1.05}{0.02857 - 0.08 + 1} = \frac{1.05}{0.94857} \approx 1.1069$$

Step 3: Solve SR equilibrium ($t = SR$)

$$(c) \quad RE \rightarrow E^e : \quad E_{SR}^e = E_{LR} = 1.05$$

$$(d) \quad UIP \rightarrow E : \quad R_{SR} = R_{SR}^* + \frac{E_{SR}^e}{E_{SR}} - 1$$

$$\Rightarrow 0.02857 = 0.08 + \frac{1.05}{E_{SR}} - 1$$

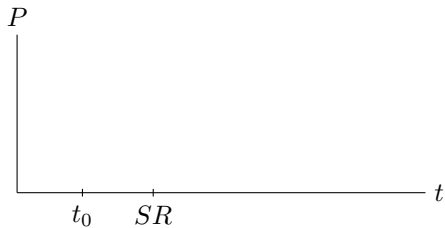
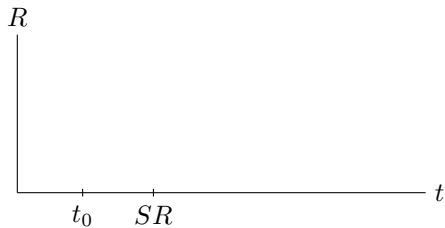
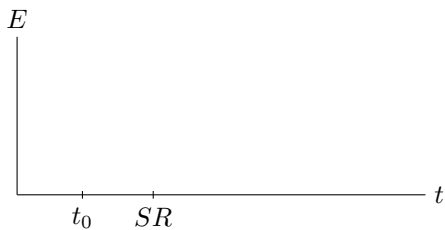
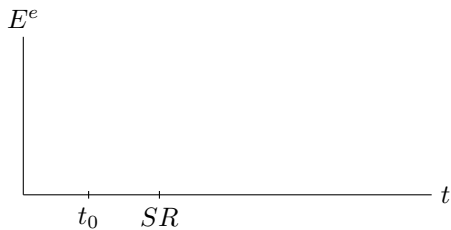
$$\Rightarrow E_{SR} = \frac{1.05}{0.02857 - 0.08 + 1} = \frac{1.05}{0.94857} \approx 1.1069$$

Solution for SR: $P_{SR} = 1.08$, $R_{SR} \approx 0.029$, $E_{SR}^e = 1.05$, $E_{SR} \approx 1.107$

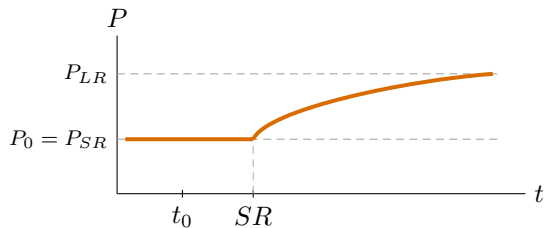
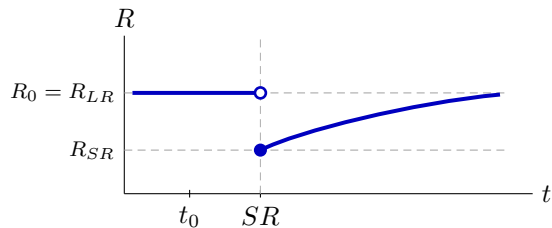
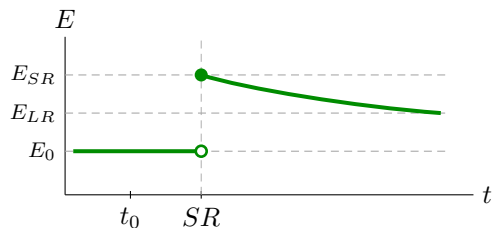
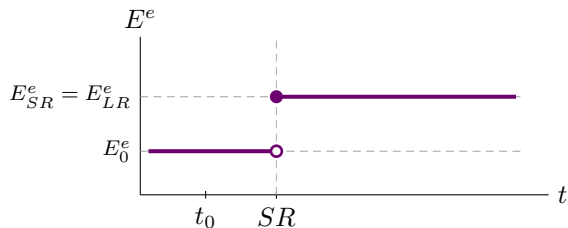
Example: solution summary

	t_0	SR	LR
P	1.08	1.08	1.134
R	0.08	0.029	0.08
E^e	1	1.05	1.05
E	1	1.107	1.05

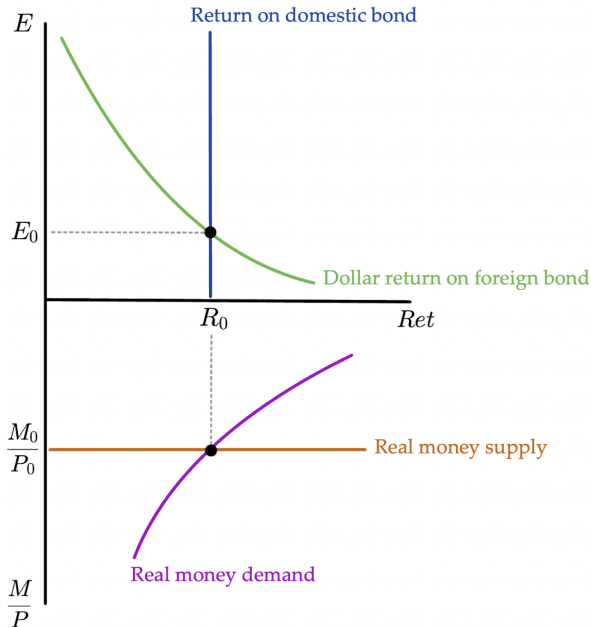
Paths of solution



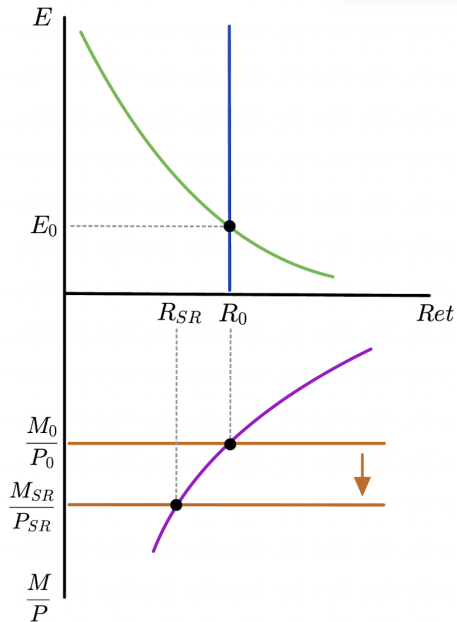
Paths of solution



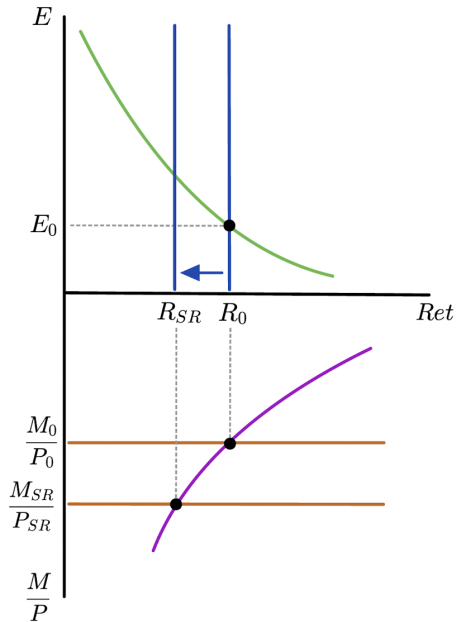
Solve with plots:
Initial LR
equilibrium



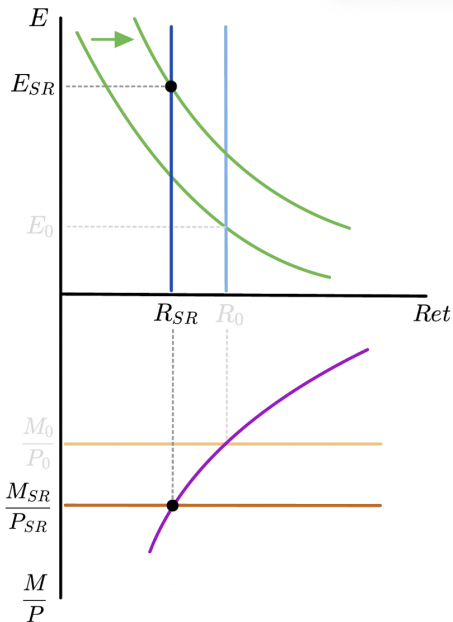
Higher M^s
lowers R



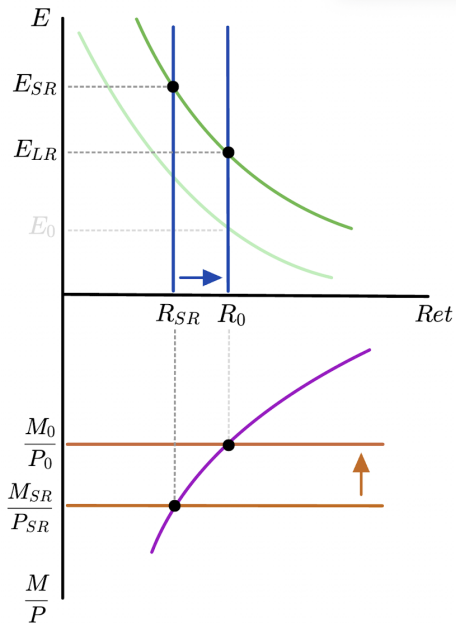
Lower R
causes
depreciation



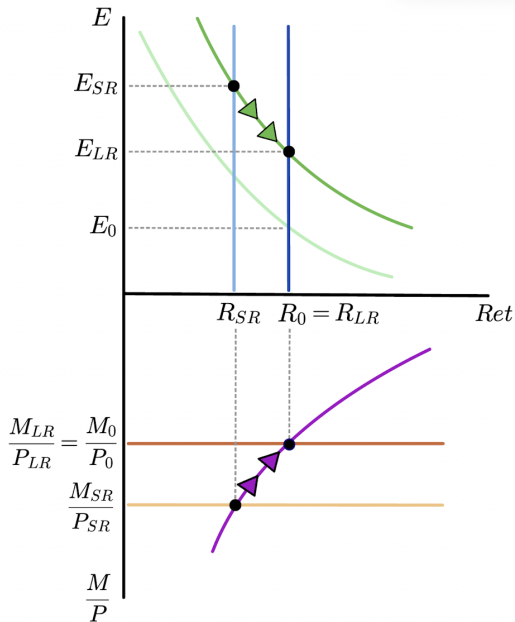
Higher E^e
shifts dollar
return on
foreign bond



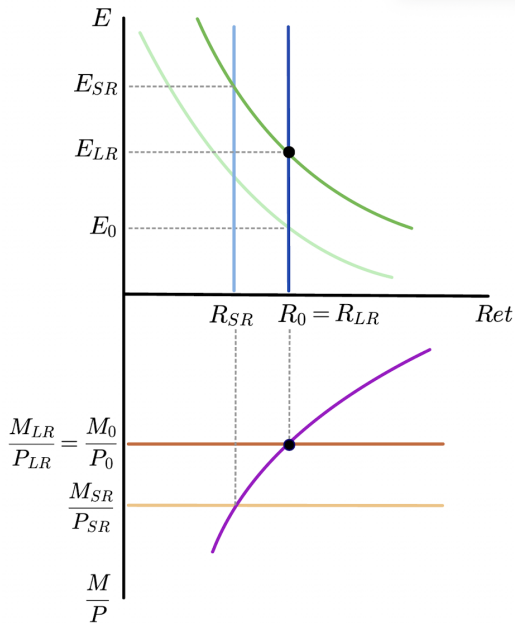
SR equilibrium



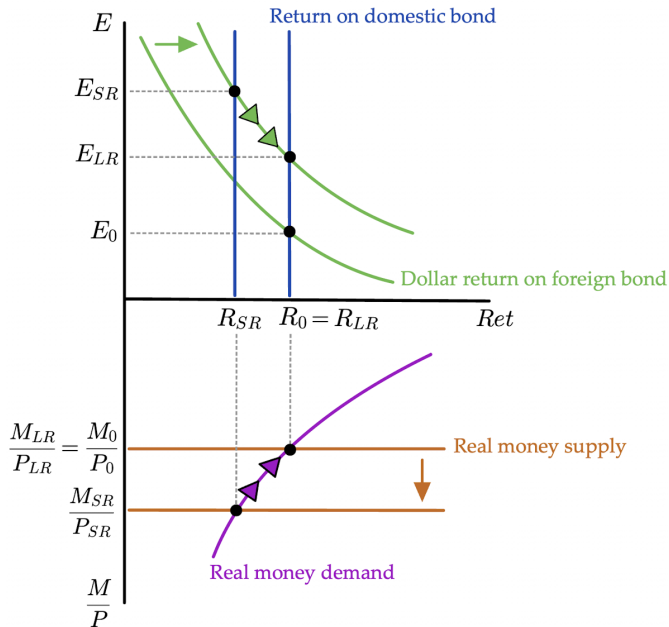
Transition between SR and LR



Final LR
equilibrium



Solve with plots:
All steps
together



Equilibrium Determination Map

	Initial	SR	LR
E	flex P / PPP	FX mkt	flex P / PPP
E^e	defn LR	RE	defn LR
R	FX mkt	money mkt	FX mkt
P	money mkt	fixed P	money mkt

PPP = purchasing power parity $\Rightarrow E \propto P \propto M^s$

defn LR = definition of the long run $\Rightarrow E_0^e = E_0$ and $E_{LR}^e = E_{LR}$

RE = rational expectations $\Rightarrow \begin{cases} E_{SR}^e = E_{LR} \\ E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR} \end{cases}$